Liniar dep. / indep.

- $v_1, \dots, v_n$ lin. dep. dc. $\exists \alpha_1, \dots, \alpha_n \in K$ (nu toți nuli) a. î.

$$\alpha_1 v_1 + \dots + \alpha_n v_n = 0$$

- $v_1, \dots, v_n$ lin. indep. dc. $\exists \alpha_1, \dots, \alpha_n \in K$ a. î.

$$\alpha_1 v_1 + \dots + \alpha_n v_n = 0 \iff \alpha_1 = \dots = \alpha_n = 0$$

Fie $A = [v_1 | v_2 | v_3]$

dc. $\operatorname{rg} A = 3 \iff A$ inversabilă $\Rightarrow v_1, v_2, v_3$ lin. indep. ($\det A \neq 0$)

dc. $\operatorname{rg} A = 2 \iff \det A = 0 \Rightarrow v_1, v_2, v_3$ lin. dep.

Valori și vectori proprii

$A \in M_n(K)$; $P_A(\lambda) = \det(A - \lambda I_n) = \begin{vmatrix} a_{11}-\lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22}-\lambda & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn}-\lambda \end{vmatrix}$

$\vec{v} \in \mathbb{R}^n$ vect. pr. pt. val. pr. λ dc. $\boxed{A \cdot \vec{v} = \lambda \cdot \vec{v}}$ $\vec{v} \neq \vec{0}$

$V_\lambda = \{ \vec{v} \mid A\vec{v} = \lambda \vec{v} \}$ subsp. vect. pr. asoc. lui λ

$\lambda_1, \dots \lambda_k \Rightarrow$ rădăcinile polinomului cu ordin de multiplicitate $m_1, \dots, m_k$

$m_1 + m_2 + \dots + m_k = n$

$\dim V_\lambda = g(\lambda) =$ multiplicitatea geometrică

T. Cayley-Hamilton

$$X^n = P_A(X) \cdot Q(X) + aX^2 + bX + c$$

Fie $A \in M_n(\mathbb{R}) \iff P_A(A) = \det(A - A \cdot I_n) = O_n$

$\Rightarrow A^n = aA^2 + bA + c \cdot I_n$

Diagonalizarea mat.

Fie $A \in M_n(\mathbb{R})$; A e diag. dc. $\exists P \in M_n(\mathbb{R})$ nesingulară ($\det P \neq 0$)
a.î. $P^{-1}AP = D$, unde D = mat. diag. $= \begin{pmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{pmatrix}$

Algoritm diag.:

1) $P_A(\lambda) = 0 = \det(A - \lambda I_n) \Rightarrow \lambda_1, \dots, \lambda_K \to n_1, \dots, n_K$

2) pt. fiecare val. proprie $\lambda \Rightarrow V_\lambda = \{\vec{v} \in \mathbb{R}^n / A\vec{v} = \lambda\vec{v}\} = Sp\{\vec{v}\}$

3) $\dim V_{\lambda_i} = n_i$; $\forall i = \overline{1,K}$ $\Rightarrow$ DA $\Rightarrow$ A se diag. $\Rightarrow$ PASUL 4
$\Rightarrow$ NU $\Rightarrow$ A nu se diag. $\Rightarrow$ END (Ghiu POWER)

4) $A = PDP^{-1}$; $D = \begin{pmatrix} \overbrace{\lambda_1 \cdots \lambda_1}^{n_1} & & 0 \\ & \ddots & \\ 0 & & \lambda_K \end{pmatrix}$; P = mat. având drept coloane vect. pr. lin. ind.

$A^n = PD^nP^{-1}$

Ortogonalizarea mat.

$u \perp v \Leftrightarrow \langle u, v \rangle = 0$

Norma : $\|u\| = \sqrt{\langle u, u \rangle}$

i) $\|u\| \geq 0$

ii) $\|u\| = 0 \Leftrightarrow u = 0$

iii) $\|\lambda \cdot u\| = \sqrt{\langle \lambda u, \lambda u \rangle} = \sqrt{\lambda^2 \langle u, u \rangle} = |\lambda| \cdot \|u\|$

$\cos(\widehat{u, v}) = \dfrac{\langle u, v \rangle}{\|u\| \cdot \|v\|}$

Prod. scalar : 1) $\langle u, v \rangle = \langle v, u \rangle$

2) $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$

3) $\langle \lambda u, v \rangle = \lambda \cdot \langle u, v \rangle$

4) $\langle u, u \rangle \geq 0$
$\langle u, u \rangle = 0 \Leftrightarrow u = 0$

$x = (x_1, \dots, x_n)$

$y = (y_1, \dots, y_n)$

$\langle x, y \rangle = \sum_{i=1}^{n} x_i \cdot y_i = x_1y_1 + x_2y_2 + \dots + x_ny_n$

$\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$

$B = \{v_1, \dots, v_n\}$ bază ortogonală dc. ⟨ bază; $v_i \perp v_j$, $\forall i \neq j$ ⇕ $\langle v_i, v_j \rangle = 0$

$B = \{v_1, \dots, v_n\}$ bază ortonormată dc. ⟨ bază ortogonală; $\|v_i\| = 1$, $\forall i$

Algoritmul Gram-Schmidt: (se aplică dc. baza B nu e ortogonală)

$B = \{v_1, v_2, v_3\}$ (bază) ⇝ $U = \{u_1, u_2, u_3\}$ (bază ortogonală) ⇝ $W = \{w_1, w_2, w_3\}$ (bază ortonormată)

$w_i = \frac{u_i}{\|u_i\|}$

$$u_1 = v_1$$

$$u_2 = v_2 - \frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} \cdot u_1$$

$$u_3 = v_3 - \frac{\langle v_3, u_1 \rangle}{\langle u_1, u_1 \rangle} \cdot u_1 - \frac{\langle v_3, u_2 \rangle}{\langle u_2, u_2 \rangle} \cdot u_2$$

$$w_1 = \frac{u_1}{\|u_1\|} \qquad w_2 = \frac{u_2}{\|u_2\|} \qquad w_3 = \frac{u_3}{\|u_3\|}$$



Complementul ortogonal al unui subsp.:

$S \subseteq V$ (subsp.); $S^\perp = \{v \mid v \perp u;\ \forall u \in S\}$

$\dim S + \dim S^\perp = \dim V$

Ortodiag. mat. simetrice

① $A \in M_n(\mathbb{R})$, $A = A^t$ ⇒ $\exists Q \in M_n(\mathbb{R})$, Q ortogonală, $\exists \Delta$ mat. diagonală a.î. $A = Q \Delta Q^t$

② $Q \in M_{m,n}(\mathbb{R})$ s.n. ortogonală dc. $\underset{m\times n}{Q} \cdot \underset{n\times m}{Q^t} = I_m$ sau $\underset{n\times m}{Q^t} \cdot \underset{m\times n}{Q} = I_n$

$Q = [q_1 | \dots | q_n] \iff \{q_1, \dots, q_n\}$ sis. ortonormat

$\|q_i\| = 1$, $\forall i$; $q_i \perp q_j$, $\forall i \neq j$

③ $A = A^t$, $\lambda_1 \neq \lambda_2$ val. proprii ⇒ $V_1 \perp V_2$ (vect. proprii) (V_1 ↙ λ_1, V_2 ↘ λ_2)

Algoritm ortodiag. mat. sim.:

I Diag. mat. A $\begin{cases} \text{val. proprii} \\ \text{vect. pr.} \\ A = P\Delta P^{-1}, \ \Delta = \begin{pmatrix} \overset{m_1}{\lambda_1 \cdots \lambda_1} & & 0 \\ & \ddots & \\ 0 & & \lambda_k \end{pmatrix}, \ P = [v_1 | \dots | v_k] \end{cases}$

$A \in M_n(\mathbb{R})$

II $\{v_1, \dots, v_k\} \xrightarrow{\text{Gram-Schmidt}} \{q_1, \dots, q_k\}$

bază — bază ortonormată

$Q = [q_1, \dots, q_k] \Rightarrow \boxed{A = Q\Delta Q^t}$

pt. că mat. e simetrică $\Rightarrow \{v_1, \dots, v_k\}$ bază ortogonală

$\Rightarrow \boxed{q = \frac{v}{\|v\|}}$

Aproximarea sol. unui sis. liniar incompatibil (pseudosoluția)

$A \in M_{m,n}(\mathbb{R})$, $b \in \mathbb{R}^m$, $b \notin Col(A)$ → spațiul coloanelor lui A

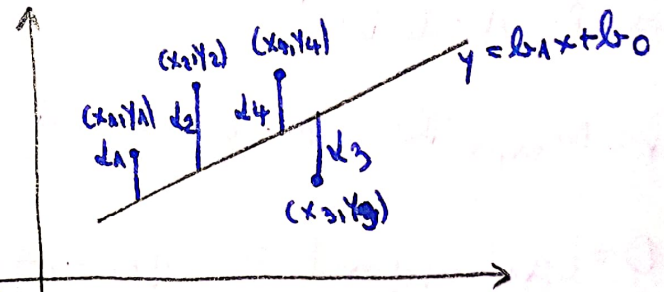
$\underset{m\times n}{A} \cdot \underset{n\times 1}{\vec{x}} = \underset{m\times 1}{\vec{b}}$ incompatibil

$proj_{ColA}\vec{b} = A \cdot x^*$, unde x^* e sol. pt. $\boxed{A^tAx = A^tb}$ sis. comp.

$Ax = b$ sis. incomp.

$\Rightarrow \boxed{x^* = \underset{n\times m}{(A^t}\underset{m\times n}{A)^{-1}}A^tb}$, $x^* = \begin{pmatrix} x_1^* \\ \vdots \\ x_n^* \end{pmatrix}$

Dreapta de regresie



$(x_1,y_1), \dots, (x_m,y_m)$

$? \begin{pmatrix} b_1 \\ b_0 \end{pmatrix}$ a.î. $y = b_1x + b_0$

$$\begin{cases} y_1 = b_1x_1 + b_0 \cdot 1 \\ y_2 = b_1x_2 + b_0 \cdot 1 \\ \dots \\ y_m = b_1x_m + b_0 \cdot 1 \end{cases} \Leftrightarrow \underbrace{\begin{pmatrix} y_1 \\ y_2 \\ \dots \\ y_m \end{pmatrix}}_{\vec{b}} = \underbrace{\begin{pmatrix} x_1 & 1 \\ x_2 & 1 \\ \dots & \dots \\ x_m & 1 \end{pmatrix}}_{A} \cdot \underbrace{\begin{pmatrix} b_1 \\ b_0 \end{pmatrix}}_{\vec{x}} \Rightarrow A \cdot \vec{x} = \vec{b} \text{ sis. incomp.}$$

$m\times 1$; $m\times 2$; 2×1

$A^tA\vec{x} = A^tb$ sis. comp.

$\boxed{\hat{x} = (A^tA)^{-1} \cdot A^tb}$

$\boxed{B^{-1} = \frac{1}{\det B}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}}$ ④ $\boxed{B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}}$

parabolă de regresie:

$y = \alpha + \beta x + \gamma x^2$

$$\begin{cases} y_1 = \alpha \cdot 1 + \beta \cdot x_1 + \gamma x_1^2 \\ y_2 = \alpha \cdot 1 + \beta \cdot x_2 + \gamma x_2^2 \\ \dots \\ y_m = \alpha \cdot 1 + \beta \cdot x_m + \gamma x_m^2 \end{cases}$$

$$\underbrace{\begin{pmatrix} y_1 \\ y_2 \\ \dots \\ y_m \end{pmatrix}}_{\vec{b}} = \underbrace{\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \dots & \dots & \dots \\ 1 & x_m & x_m^2 \end{pmatrix}}_{A} \cdot \underbrace{\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}}_{\vec{x}}$$

$$\boxed{\vec{x} = (A^t A)^{-1} \cdot A^t \vec{b} = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}}$$

Factorizare QR

$A \in M_{m,n}(\mathbb{R}) \Rightarrow \underset{m\times n}{A} = \underset{m\times m}{Q}\,\underset{m\times n}{R}$

- Q ortogonală: $QQ^t = I_m$; $Q^tQ = I_m$
- R superior triunghiulară

$$R = \begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{pmatrix}$$

$\{v_1, \dots, v_n\}$ lin. ind.

$Q = [q_1 | \dots | q_n]$

$A = [v_1 | \dots | v_n] \Rightarrow \{v_1, \dots, v_n\} \xrightarrow{G\text{-}S} \{q_1, \dots, q_n\}$

Exemplu: $R \in M_3(\mathbb{R}) \Rightarrow R = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \end{pmatrix}$

$r_{11} = \langle v_1, q_1 \rangle$, $r_{12} = \langle v_1, q_2 \rangle$, $r_{13} = \langle v_1, q_3 \rangle$

$r_{22} = \langle v_2, q_2 \rangle$, $r_{23} = \langle v_2, q_3 \rangle$, $r_{33} = \langle v_3, q_3 \rangle$

Descompunerea QR pt. rezolvarea sis. incomp.: $\boxed{R\vec{x} = Q^t \vec{b}}$

Forme pătratice. Conice

Fie $A \in M_n(\mathbb{R})$ o mat. simetrică.

S.n. formă pătratică asoc. mat. A funcția $g: \mathbb{R}^n \to \mathbb{R}$, $g(v) = v^t A v$

Notăm $v = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ și $A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix}$

$$\Rightarrow g(v) = (x_1 \dots x_n) \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \sum_{i,j=1}^{n} a_{ij} x_i x_j$$

Extremele formelor pătratice:

Fie $A \in M_n(\mathbb{R})$ matr. sim. și $g: \mathbb{R}^n \to \mathbb{R}$, $g(v) = v^t A v$ forma pătratică asoc.

Dc. $\{v_1, \dots, v_n\}$ bază ortonormată pt. $\mathbb{R}^n$ cu vectori proprii ai matr. A, coresp. val. proprii $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ (în această ordine), at.:

1) $\max\limits_{\|v\|=1} g(v) = g(v_1) = \lambda_1$

2) $\min\limits_{\|v\|=1} g(v) = g(v_n) = \lambda_n$

Exemplu: $g\begin{pmatrix} x \\ y \end{pmatrix} = \underline{5}x^2 + \underline{4}xy + \underline{8}y^2 \Rightarrow A = \begin{pmatrix} \underline{5} & \underline{2} \\ \underline{2} & \underline{8} \end{pmatrix}$

O formă pătratică $g: \mathbb{R}^n \to \mathbb{R}$, $g(v) = v^t A v$ este pozitiv definită dc. $g(v) > 0$, $\forall v \in \mathbb{R}^n$, $v \neq 0$.

O formă pătratică este negativ definită dc. $g(v) < 0$, $\forall v \in \mathbb{R}^n$, $v \neq 0$.

O formă pătratică este nedef. dc. nu e nici poz. nici neg. def.

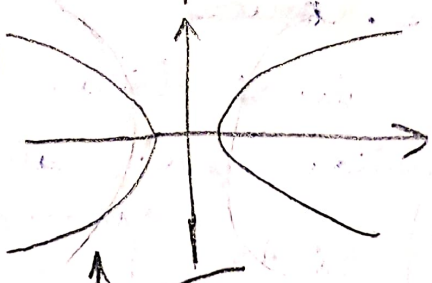
——"—— poz. def. (resp. neg. def.) $\Leftrightarrow$ toate val. proprii ale matr. A sunt strict poz. (resp. neg.)

Tipuri de conice:

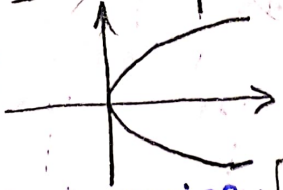
1) Elipsă: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



2) Hiperbolă: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$



3) Parabolă: $y^2 = 2px$



Ecuația generală a unei conice: $\boxed{ax^2 + 2bxy + cy^2 + dx + ey + f = 0}$

unde $a, b, \dots \in \mathbb{R}$, cu $a^2 + b^2 + c^2 \neq 0$

Exemplu: Fie conica $ax^2+2bxy+cy^2+dx+ey+f=0$

Matricial devine: $(x\ y)\underbrace{\begin{pmatrix} a & b \\ b & c \end{pmatrix}}_{A}\begin{pmatrix} x \\ y \end{pmatrix}+\underbrace{(d\ \ e)}_{\vec{b}}\begin{pmatrix} x \\ y \end{pmatrix}+f=0$

Dc. ortodiag. matr. A: $A=Q\Delta Q^t$, unde $\Delta=\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$ $\Rightarrow$

$\Rightarrow (x\ y)\,Q\Delta Q^t\begin{pmatrix} x \\ y \end{pmatrix}+(d\ \ e)\begin{pmatrix} x \\ y \end{pmatrix}+f=0$

Notăm $\begin{pmatrix} x' \\ y' \end{pmatrix}=Q^t\begin{pmatrix} x \\ y \end{pmatrix}$ $\Rightarrow$

$\Rightarrow (x'\ y')\,\Delta\begin{pmatrix} x' \\ y' \end{pmatrix}+(d\ \ e)\,Q\begin{pmatrix} x' \\ y' \end{pmatrix}+f=0 \iff$

$\iff \lambda_1 x'^2+\lambda_2 y'^2+d'x'+e'y'+f=0$

Grupăm în pătrate perfecte: $\lambda_1(\underbrace{x'+\dots}_{x''})^2+\lambda_2(\underbrace{y'+\dots}_{y''})^2=\dots$

Și obținem: $\lambda_1 x''^2+\lambda_2 y''^2=\dots$

ec. conicei în forma canonică

Descompunerea în val. singulare (SVD)

$A\in M_{m,n}(\mathbb{R}) \Rightarrow \exists\ U\in M_m(\mathbb{R}),\ V\in M_n(\mathbb{R})$ ortogonale

$U\cdot U^t=I_m$

$V\cdot V^t=I_n$

$\Rightarrow \exists\ \sigma_1\geq\sigma_2\geq\dots\geq\sigma_r>0$ a.î.

$$\underset{m\times m}{U^t}\ \underset{m\times n}{A}\ \underset{n\times n}{V}=\left(\begin{array}{c|c} \begin{matrix}\sigma_1 & & 0\\ & \ddots & \\ 0 & & \sigma_r\end{matrix} & 0 \\ \hline 0 & 0 \end{array}\right)\begin{matrix}\}r\text{ linii}\\ \}m-r\text{ linii}\end{matrix}=\underset{m\times n}{\Sigma}$$

(r col. | $n-r$ col.)

$\Rightarrow \boxed{A=U\Sigma V^t}$

$\sigma_1,\dots,\sigma_r$ val. singulare

Algoritm det. SVD $\Rightarrow A=U\Sigma V^t$

① Se ortodiag. matr. sim. $B=\underset{n\times m}{A^t}\underset{m\times n}{A}$ $\Rightarrow V: n\times n$

$B^t=B$

$B=Q\Delta Q^t$; $Q=[v_1|\dots|v_n]=V$

$v_1,\dots,v_n$ ortonormați

Q ortogonală

$\lambda_1\geq\lambda_2\geq\dots\geq\lambda_r>0$; $\lambda_{r+1}=\dots=\lambda_n=0$; $r=\operatorname{rg}A=\operatorname{rg}A^t$

② $\boxed{\sigma_i = \sqrt{\lambda_i} \;;\; i = \overline{1,r}} \Rightarrow \Sigma = \left(\begin{array}{c|c} \operatorname{diag}(\sigma_1, \dots, \sigma_r) & 0 \\ \hline 0 & 0 \end{array}\right)$

$V = (v_1 | \dots | v_n)$

③ $U = (u_1 | \dots | u_r | u_{r+1} | \dots | u_m)$

$\boxed{u_i = \frac{1}{\sigma_i} A v_i \;;\; i = \overline{1,r}} \Rightarrow \{u_1, \dots, u_r\}$ ortonormați $\xrightarrow{\text{completăm}} \{u_1, \dots, u_m\}$ bază ortonormată în $\mathbb{R}^m$

Ecuații diferențiale liniare de ord. I

$y'(x) = f(x) \cdot y + g(x)$

$\boxed{y_o(x) = K \cdot e^{\int f(x)\,dx}}$ $\qquad$ $\boxed{y_p(x) = e^{\int f(x)\,dx} \cdot \int g(x) e^{-\int f(x)\,dx}\,dx}$

$\boxed{y(x) = y_o(x) + y_p(x) = e^{\int f(x)\,dx}\left(c + \int g(x) e^{-\int f(x)\,dx}\,dx\right),\ c \in \mathbb{R}}$

Verificare: - se derivează $y_p(x)$ rezultat
- se înlocuiește în ec. $y'(x) = f(x) \cdot y + g(x)$ astfel:
$y_p'(x) = f(x) \cdot y_p(x) + g(x)$
- se compară cele două rezultate

Ec. dif. liniare de ordin n cu coef. const.

$a_0 y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y' + a_n y = f(x)$

$y(x) = y_o(x) + y_p(x)$

I Ec. omogenă: $a_0 y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y' + a_n y = 0$

$a_0 \lambda^n + a_1 \lambda^{n-1} + \dots + a_{n-1}\lambda + a_n = 0$ (ec. caract. asoc. E.D.)

ⓘ $\lambda_1, \lambda_2, \dots, \lambda_n$ rād. reale dist. $\Rightarrow e^{\lambda_1 x}, e^{\lambda_2 x}, \dots, e^{\lambda_n x}$ sol. lin. ind. pt. E.D.O.

$\Rightarrow \boxed{y_o(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} + \dots + c_n e^{\lambda_n x}}$

ⓘⓘ $\lambda_1 = \lambda_2 \in \mathbb{R}$; $\lambda_3, \dots, \lambda_n$ rād. reale dist. $\Rightarrow e^{\lambda_1 x}, x e^{\lambda_1 x}, e^{\lambda_3 x}, \dots, e^{\lambda_n x}$ sol.

$\Rightarrow \boxed{y_o(x) = c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_1 x} + c_3 e^{\lambda_3 x} + \dots + c_n e^{\lambda_n x}}$

(iii) $\lambda_1 = a+ib$
$\lambda_2 = a-ib$; $\lambda_3, \dots, \lambda_n \in \mathbb{R}$ dist.

$g(x) = e^{\lambda x} = e^{(a+ib)x} = e^{ax} \cdot e^{ibx} = e^{ax}(\cos bx + i \sin bx)$

$g(x) = e^{ax}\cos bx + i e^{ax} \sin bx \Rightarrow \operatorname{Re} g(x)$; $\operatorname{Im} g(x)$ sol. reale pt. E.O.

$\Rightarrow \boxed{y_o(x) = c_1 e^{ax}\cos bx + c_2 e^{ax}\sin bx + c_3 e^{\lambda_3 x} + \dots + c_n e^{\lambda_n x}}$

II Ec. neomogenă : $a_0 y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y' + a_n y = f(x)$

Cazuri particulare pt. term. liber $f(x)$:

1) $f(x) = A e^{\alpha x} \Rightarrow$ (i) $\boxed{y_p(x) = a e^{\alpha x}}$ dc. α nu e rad. a ec. caract.
$\Rightarrow$ (ii) $\boxed{y_p(x) = a x^k e^{\alpha x}}$ dc. α e rad. a ec. caract. cu multiplicitatea k
(sol. pt. ec. omog. $\rightarrow e^{\alpha x}, x e^{\alpha x}, \dots, x^{k-1} e^{\alpha x}$)

2) $f(x) =$ polinom de grad $m \Rightarrow \boxed{y_p(x) =$ polinom de același grad cu $f}$

3) $f(x) = a\cos \lambda x + b \sin \lambda x \Rightarrow \boxed{y_p(x) = A\cos \lambda x + B \sin \lambda x}$

4) $f(x) = P_m(x) e^{\alpha x} \Rightarrow \boxed{y_p(x) = x^k Q_m(x) e^{\alpha x}}$ dc. α e rad. a ec. caract. de mult. k

III $y(x) = y_o(x) + y_p(x)$

Pentru sisteme de ec. dif. vezi seminarul 14.